Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/15/2025
PI Name:	Jain, Abhishek Ph.D.		
Project Title:	Long-term Patient iPSC Vessel Chip Model to Assess Stressors of Atherosclerosis and mRNA Therapeutics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	a.jain@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-458-8494
Organization Name:	Texas A&M Engineering Experiment Station		
PI Address 1:	101 Bizzell St, ETB 5045		
PI Address 2:	3120 TAMU		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	03/20/2023	**End Date:**	02/20/2027
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	NA		
COI Name (Institution):	Le, Nhat Tu Ph.D. (Houston Methodist Hospital) Yakovlev, Vladislav Ph.D. (Texas A&M Engineering Experiment Station) Wang, Guangyu Ph.D. (Houston Methodist Hospital) Cooke, John P M.D., Ph.D. (The Houston Methodist Research Institute)		
Grant/Contract No.:	80ARC023CA002		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Animals are not very precise modeling systems of human physiology and cannot ensure dissectible signaling analysis. Recent advances in organ-on-chip technology or microphysiological systems (MPS) have permitted co-culture of human cells in physiologically-relevant microfluidic environments and also allowed continuous perfusion, providing an alternative in vitro approach to mimic the human disease. These microsystems, however, still have not reached their full potential and made in-depth entry into informing clinical trials due to the absence of efficient and scalable primary human patient cells, limited real-time biosensing potential, and critical validation over long durations. Amongst all human disease conditions, vascular diseases can be often fatal in which symptoms such as aggressive atherosclerosis exist. Amongst other risk factors, age is one of the most critical. Radiation exposure during space travel is another. Therefore, investigating vascular signaling mechanisms and finding therapeutics for aging and radiation-related vascular diseases is of high prospective impact. But there is a fundamental gap in the understanding of: 1) how organ-level signaling occurs in blood vessels amongst atherosclerotic patients, and 2) how cells inside the vascular organ contribute to the disease and respond to drugs over time. To meet these complex challenges, the team's goal is to take Progeria patients and proton radiation as case-studies and demonstrate that 3D cylindrical vessel-chips containing co-culture of vascular endothelial and smooth muscle cells from human induced pluripotent stem cells (hiPSCs) will provide real-time readouts of critical vascular and hematological biomarkers, that inform clinical outcomes. The chip will consist of physiologically-relevant extracellular matrix that will be continually monitored and it will also be ready for microgravity experiments that can be done in space. The objective is to establish long-term culture and feasibility of a Vessel-Chip system that helps dissect pathophysiology of atherosclerosis and toxin/drug-tissue interactions with much higher specificity and sensitivity than gold-standard models. We will add more advanced biosensing to our existing Vessel-Chip that mimics vessel geometry, hemodynamics, and extracellular matrix, to continuously monitor physiological parameters and sustain the culture of human iPSC-derived endothelial cells (hiPSC-EC) and vascular smooth muscle cells (hiPSC-SMC) for 180 days from healthy patients (Phase I). We will assess HGPS patient iPSC vascular cells and protons as biological and space-relevant stressors respectively over 180 days, and finally, novel Good Manufacturing Practice (GMP)-grade mRNA therapeutics loaded in a unique lipid nanoparticle-based delivery vehicle will be evaluated through a panoply of organ-to-gene level assays, including clinically-relevant biomarkers (Phase II). The research proposed is innovative and significant, because it contributes in shifting the status quo of vascular pathophysiology in the organ-on-a-chip platform by containing primary hiPSC-vascular cells, blood flow, and associated transcriptomics and proteomics over long durations of co-culture and continuous monitoring. The outcomes will lead to further applications of the organ-on-chip technology in translational research in atherosclerosis and other vascular and hematological diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: -- A long-duration microphysiological model of atherosclerosis. -- New knowledge of aging and radiation-induced vascular disease. -- A testbed for mRNA therapeutic preclinical trials. -- Enabling future experimentation at the International Space Station (ISS) under microgravity.
Task Progress:	The team assembled to execute this project has expertise in a) organ-on-chip engineering and innovation; b) vascular aging and stem cell biology; and c) RNA drug development against HGPS (Hutchinson-Gilford Progeria Syndrome). From this current ongoing NASA-led multi-agency contract, our team has accomplished the following: 1. Created living arterial vessel-chips consisting of normal endothelial cells under flow that surpass 180 days of culture. We can co-culture smooth muscle cells and surpass 145 days of culture (ongoing), and we can introduce RNA-therapeutics in these model systems. Therefore, we have almost achieved the milestone of Phase 1 of this project. Additionally, we have: 2. Characterized the cellular and transcriptional changes over time of vascular cells on the chips using cutting-edge transcriptomics. 3. Designed and successfully tested a novel circular RNA therapeutic that reverses cellular aging, and developed cGMP processes for generating clinical grade linear and circular mRNA. 4. Engineered Hemadyne, a perfusion system capable of introducing multiphasic clinical hemodynamic waveforms in vessel chips, including capability to recreate flow reversal patterns as seen in patients and astronauts using doppler imaging. 5. Designed a machine learning (ML)-based analytical pipeline to automate and rapidly assess endothelial health in vessel chips using simple real-time brightfield image acquisition needed in this proposal. 6. We have engineered a wireless sensor and monitoring device that provides real-time readouts of temperature, humidity, and Co2 within cell culture systems, essential to maintain the quality of long-term cultures.
Bibliography Type:	Description: (Last Updated: 06/02/2026)
Articles in Peer-reviewed Journals	Lee JD, Kumar A, Mathur T, Jain A. "Vascular architecture-on-chip: Engineering complex blood vessels for reproducing physiological and heterogeneous hemodynamics and endothelial function." Lab Chip. 2025 May 28;25(11):2620-31. https://doi.org/10.1039/d4lc00968a ; PubMed PMID: 40067315; PubMed Central PMCID: PMC11895859 , May-2025
Articles in Peer-reviewed Journals	Ghosh LD, Jain A. "The prospects of microphysiological systems in modeling platelet pathophysiology in cancer." Platelets. 2023 Dec 31;34(1):2247489. Review. https://doi.org/10.1080/09537104.2023.2247489 ; PMID: 37610007; PMCID: PMC10578702 , Dec-2023
Articles in Peer-reviewed Journals	Tronolone JJ, Mohamed N, Chaftari CP, Sun Y, Mathur T, Jain A. "Engineering and evaluating vascularized organotypic spheroids on-chip." Current Protocols. 2024 Nov 21;4(11):e70058. https://doi.org/10.1002/cpz1.70058 ; PMID: 39570194 , Nov-2024

Articles in Peer-reviewed Journals	Cooke JP. "From the masters: How RNA drugs will transform vascular medicine." Vasc Med. 2025 Apr;30(2):120-3. https://doi.org/10.1177/1358863X251323275 ; PMID: 40028684 , Apr-2025
Articles in Peer-reviewed Journals	Abe J, Chau K, Mojiri A, Wang G, Oikawa M, Samanthapudi VSK, Osborn AM, Ostos-Mendoza KC, Mariscal-Reyes KN, Mathur T, Jain A, Herrmann J, Yusuf SW, Krishnan S, Deswal A, Lin SH, Kotla S, Cooke JP, Le N-T. "Impacts of radiation on metabolism and vascular cell senescence." Antioxid Redox Signal. 2025 Jul;43(1-3):92-114. Review. https://doi.org/10.1089/ars.2024.0741 ; PubMed PMID: 40233257 , Jul-2025
Articles in Peer-reviewed Journals	Harrington J, Cheburkanov V, Kizilov M, Kulagin I, Petrov GI, Yakovlev VV. "Highly sensitive, low-cost deep-UV resonant Raman microspectroscopy systems." Chemistry-Methods. 2025 Sep;5(9):202500006. https://doi.org/10.1002/cmtd.202500006 , Sep-2025
Articles in Peer-reviewed Journals	Coker ZN, Troyanova-Wood M, Steelman ZA, Ibey BL, Bixler JN, Scully MO, Yakovlev VV. "Brillouin microscopy monitors rapid responses in subcellular compartments." PhotoniX. 2024 Apr 10;5(1):9. https://doi.org/10.1186/s43074-024-00123-w ; PMID: 38618142; PMCID: PMC11006764 , Apr-2024
Articles in Peer-reviewed Journals	Cheburkanov V, Du J, Brogan DM, Berezin MY, Yakovlev VV. "Toward peripheral nerve mechanical characterization using Brillouin imaging spectroscopy." Neurophotonics. 2023 Aug 26;10(3):035007. https://doi.org/10.1117/1.NPh.10.3.035007 ; PMID: 37635849; PMCID: PMC10460255 , Aug-2023
Articles in Peer-reviewed Journals	Vasanthakumari P, Romano RA, Rosa RGT, Salvio AG, Yakovlev V, Kurachi C, Hirshburg JM, Jo JA. "Pixel-level classification of pigmented skin cancer lesions using multispectral autofluorescence lifetime dermoscopy imaging." Biomed Opt Express. 2024 Jul 9;15(8):4557-83. https://opg.optica.org/boe/abstract.cfm?URI=boe-15-8-4557 ; PMID: 39346997; PMCID: PMC11427192 , Jul-2024
Articles in Peer-reviewed Journals	Li T, Cheburkanov V, Yakovlev VV, Agarwal GS, Scully MO. "Harnessing quantum light for microscopic biomechanical imaging of cells and tissues." Proc Natl Acad Sci. 2024 Sep 24;121(45):e2413938121. https://doi.org/10.1073/pnas.2413938121 ; PMID: 39480851; PMCID: PMC11551316 , Sep-2024
Articles in Peer-reviewed Journals	Cheburkanov V, Jung S, Kizilov M, Holt SE, Alge DL, Ware TH, Yakovlev VV. "Brillouin spectroscopy: A non-invasive method for assessing the viscoelastic properties of biologically relevant polymers." J Biomed Mater Res A. 2025 Jul 13;113(7):e37965. https://doi.org/10.1002/jbm.a.37965 ; PMID: 40653450 , Jul-2025
Articles in Peer-reviewed Journals	Abasi S, Jain A, Cooke JP, Guiseppi-Elie A. "Electrically stimulated gene expression under exogenously applied electric fields." Frontiers in Molecular Biosciences. 2023 May 4;10:1161191. https://doi.org/10.3389/fmolb.2023.1161191 , May-2023
Articles in Peer-reviewed Journals	Willis JA, Trevino A, Nguyen C, Benjamin CC, Yakovlev VV. "Photodynamic therapy minimally affects HEMA-DMAEMA hydrogel viscoelasticity. " Macromolecular bioscience. 2023 Nov;23(11):2300124. https://doi.org/10.1002/mabi.202300124 , Nov-2023
Articles in Peer-reviewed Journals	Cheburkanov V, Keene E, Pipal J, Johns M, Applegate BE, Yakovlev VV. "Porcine vocal fold elasticity evaluation using Brillouin spectroscopy. " Journal of Biomedical Optics. 2023 Aug 1;28(8):087002. https://doi.org/10.1117/1.JBO.28.8.087002 , Aug-2023
Articles in Peer-reviewed Journals	O'Connor SP, Doktor DA, Scully MO, Yakovlev VV. "Spectral resolution enhancement for impulsive stimulated Brillouin spectroscopy by expanding pump beam geometry." Optics express. 2023 Apr 18;31(9):14604-16. https://doi.org/10.1364/OE.487131 , Apr-2023